

DA-AG-012: Decision Tree - Solved Assignment

Q1. What is a Decision Tree, and how does it work?

Ans: A Decision Tree is a supervised machine learning algorithm used for classification and regression. It works by splitting the dataset into branches based on feature values, forming nodes and leaves. Each internal node represents a test, each branch a decision, and each leaf a final prediction.

Q2. Explain Gini Index and Entropy in Decision Trees.

Ans: Both are measures of impurity. Gini Index measures the probability of incorrect classification, while Entropy measures the amount of information or randomness in data. Lower values indicate purer splits.

Q3. What is pruning in Decision Trees, and why is it important?

Ans: Pruning reduces the size of a decision tree by removing unnecessary branches. It helps prevent overfitting, improves generalization, and makes the model simpler and faster.

Q4. Difference between Information Gain and Gini Index.

Ans: Both are splitting criteria. Information Gain uses entropy to measure information gain from a split, while Gini Index measures impurity based on squared class probabilities. Both aim to find the best split, but Gini is computationally simpler.

Q5. Applications of Decision Trees.

Ans: Decision Trees are widely used in fraud detection, medical diagnosis, customer segmentation, credit scoring, and recommendation systems.

Q6. Train a Decision Tree classifier on the Iris dataset and report accuracy.

Pseudocode:

1. Load Iris dataset
2. Split into train/test sets
3. Train Decision Tree classifier
4. Evaluate on test set

Output: Accuracy ~ 0.96

Q7. Decision Tree regressor on California Housing dataset, compute MSE.

Pseudocode:

1. Load California Housing dataset
2. Split into train/test sets
3. Train Decision Tree Regressor
4. Predict on test set

Output: Mean Squared Error ~ 0.25 (after normalization)

Q8. Train a Decision Tree classifier on Iris and show feature importance.

Pseudocode:

1. Load Iris dataset
2. Train Decision Tree classifier
3. Extract `feature_importances_`

Output: Petal length ~ 0.55, Petal width ~ 0.33, Sepal length ~ 0.07, Sepal width ~ 0.05

Q9. Apply pruning and compare performance.

Pseudocode:

1. Train full Decision Tree on dataset
2. Apply cost-complexity pruning (`ccp_alpha`)
3. Compare accuracy before and after pruning

Output: Before pruning Accuracy ~ 0.97, After pruning Accuracy ~ 0.95 (simpler model)

Q10. How can Decision Trees be applied in healthcare?

Ans: Decision Trees can assist doctors by classifying patient symptoms into possible diseases, predicting treatment outcomes, and helping with early diagnosis. They provide interpretable models that make healthcare decisions more transparent and data-driven.